

Lab 1 — Suricata Fundamentals

➤ Learning Objectives

By the end of this lab, students should be able to:

- Explore the **Suricata environment**.
 - Analyze and validate a **basic Suricata configuration**.
 - Understand how Suricata processes **network traffic and PCAP files**.
 - Load and test **intrusion detection rulesets**.
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➤ Prerequisites

- Suricata is installed and running on your Linux machine (Ubuntu Desktop or Server recommended).
 - You must have access to a terminal and sudo privileges.
 - You must have test PCAP files (you can create or download examples).
 - Official Suricata documentation: <https://docs.suricata.io/>
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➤ Lab Scenario

You are a Blue Team analyst responsible for deploying Suricata on a monitoring system. Your goal is to verify the installation, inspect the configuration, load detection rules, and ensure the IDS is ready to monitor network traffic.

Complete the following missions and provide the required deliverables.

Mission 1: Verify Suricata Installation

Your first task is to verify that Suricata is correctly installed and operational.

Tasks:

- Verify that Suricata is installed on your system.
- Check whether the Suricata service is running.
- Identify the installed **Suricata version**.
- Locate the **main configuration file** used by Suricata.

Mission 2: Explore the Configuration

Suricata relies heavily on its configuration file.

Tasks:

Locate and analyze the **suricata.yaml** configuration file.

Identify the following elements:

- The **HOME_NET** variable
- The directory containing **rules**
- The directory where **logs** are stored

Modify the **HOME_NET** variable so that it matches your **local network range**.

Mission 3: Define Network Sources

Suricata must know which network interface to monitor.

Tasks :

- Identify all network interfaces available on your system.
- Select the interface that carries the **main network traffic**.
- Configure Suricata to listen on this interface.

Mission 4: Add a Ruleset

An IDS without detection rules cannot identify threats.

Tasks :

- Download an **open-source Suricata ruleset**.
- Add the ruleset to the Suricata configuration.
- Ensure that Suricata loads the rules successfully.

Mission 5: Validate the Configuration

Before running Suricata in monitoring mode, the configuration must be tested.

Tasks :

- Test the configuration file using the appropriate Suricata command.
- Identify and fix any configuration errors.
- Confirm that the configuration is valid.

Deliverables

Provide:

- The installed Suricata version
- The original value of **HOME_NET**
- The modified value
- A brief explanation of your change
- The selected network interface
- A short justification for your choice
- The number of rules loaded
- A screenshot or output showing a successful configuration test

Tips & Best Practices

- Always work on a **copy of the configuration file** to avoid breaking the main Suricata instance.
- When testing detection capabilities, start with a **small PCAP file** before analyzing live traffic.
- **Carefully read error messages produced during configuration testing**; they usually provide clear guidance for troubleshooting.

Reflection Questions

- What is the role of a **ruleset** in an Intrusion Detection System?
- Why is it important to test a Suricata configuration before running it?
- What is the difference between **HOME_NET** and **EXTERNAL_NET**?
- Why does Suricata require a well-defined configuration file before capturing traffic?
- What is the difference between running Suricata with the **default configuration** and running it with **custom rules**?

“Every alert is a clue; your network never sleeps!”